PRAKTIKUM BASIS DATA

MODUL 8

Scripting_SQL_Create

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

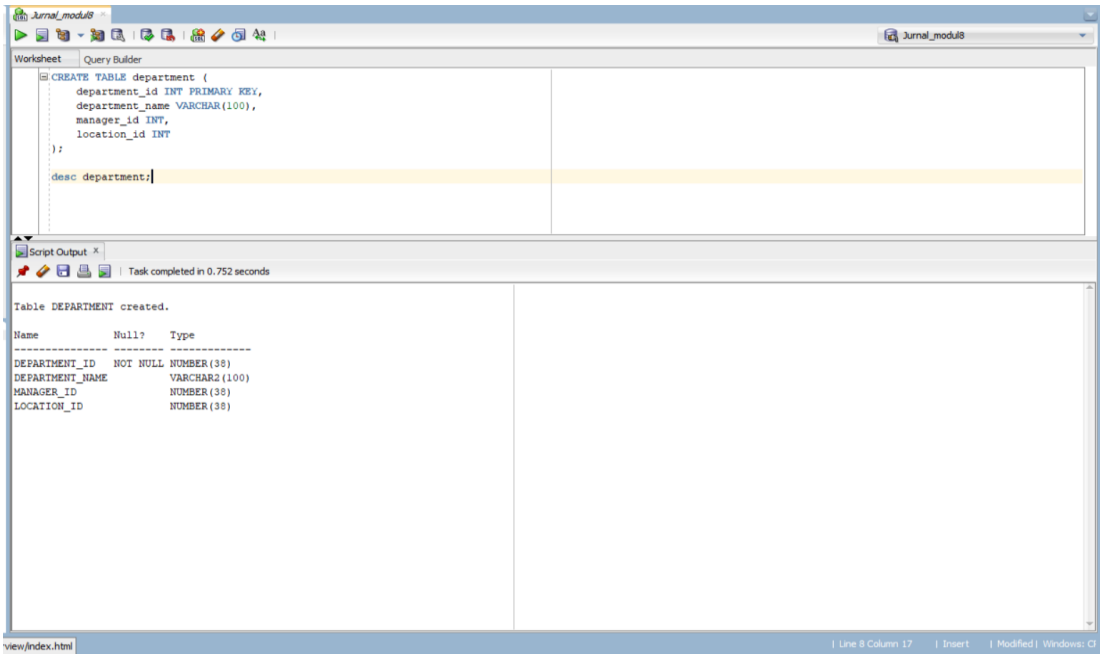
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.



b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>

```
CREATE TABLE employee (
    employee_id INT PRIMARY KEY,
    last_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    salary DECIMAL(10,2),
    commission_pct DECIMAL(5,2),
    hire_date DATE NOT NULL,
    department_id INT,
    FOREIGN KEY (department_id) REFERENCES department(department_id)
);

desc employee;
```

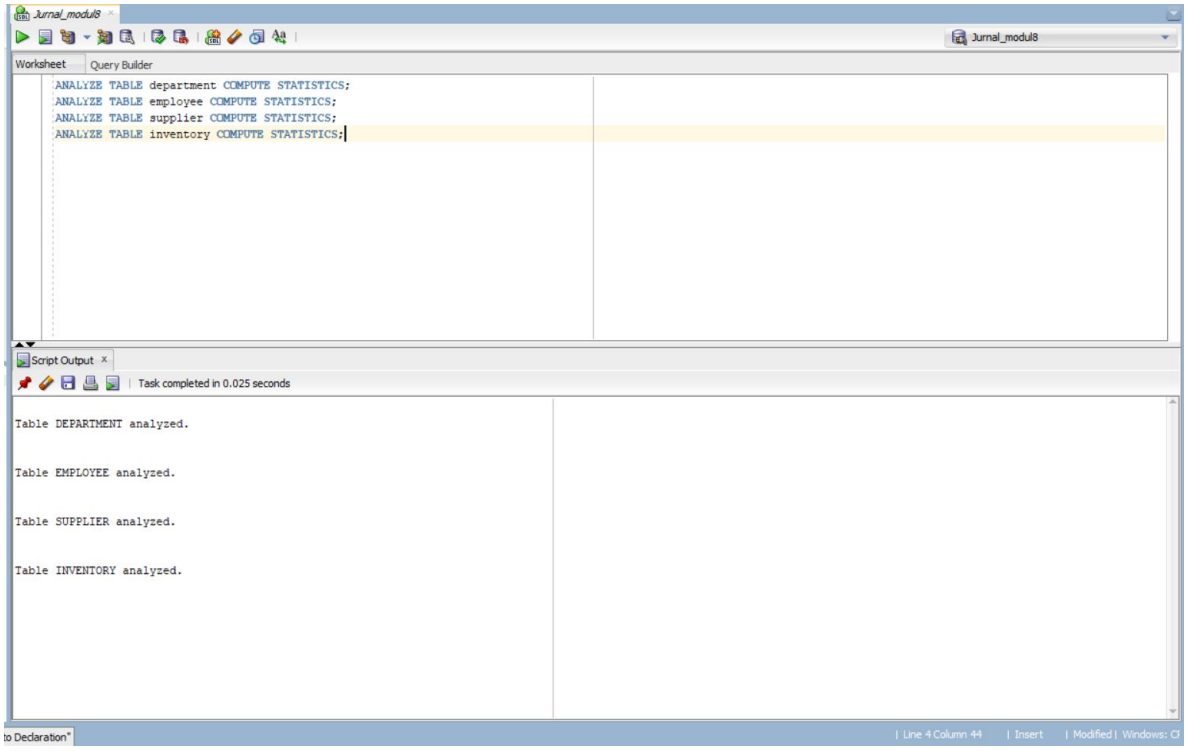
```
Task completed in 0.342 seconds

Table EMPLOYEE created.

Name           Null?    Type
-------------- -------- -------------
EMPLOYEE_ID    NOT NULL NUMBER(38)
LAST_NAME      NOT NULL VARCHAR2(100)
EMAIL                   VARCHAR2(100)
SALARY                  NUMBER(10,2)
COMMISSION_PCT          NUMBER(5,2)
HIRE_DATE      NOT NULL DATE
DEPARTMENT_ID           NUMBER(38)
```

c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst)

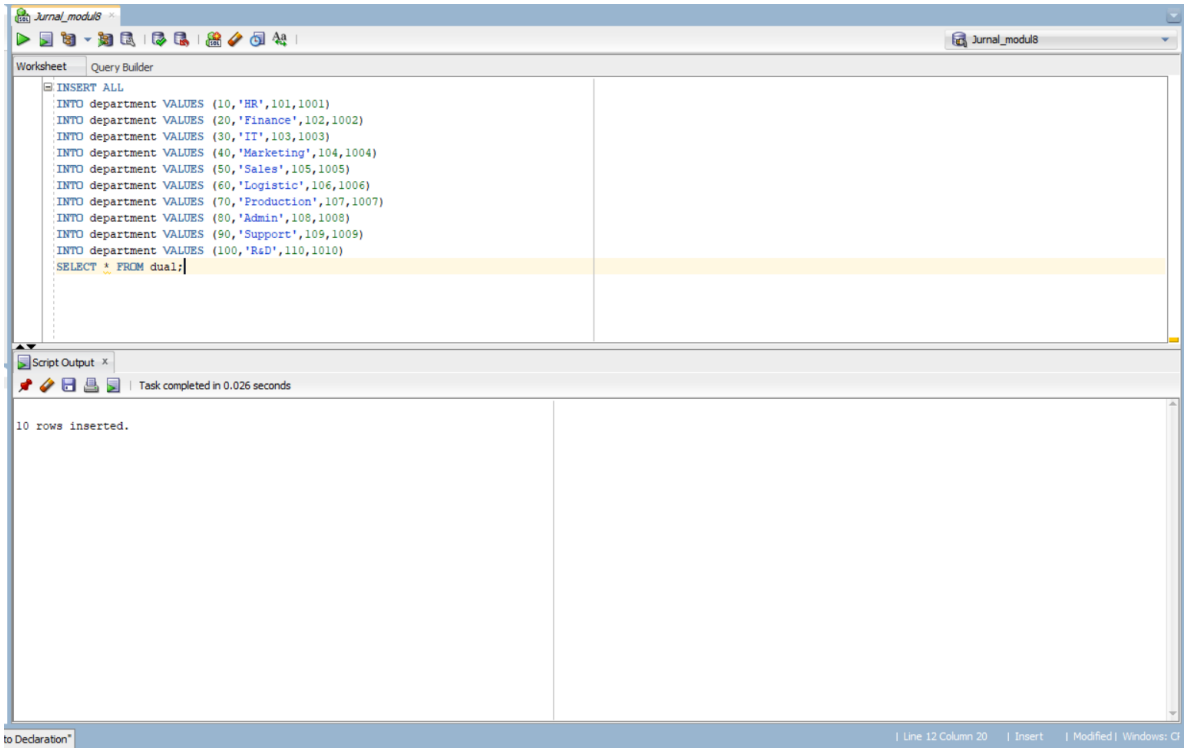
```
CREATE TABLE supplier (
    supplier_id INT PRIMARY KEY,
    supplier_name VARCHAR(100),
    contact VARCHAR(100)
);

CREATE TABLE inventory (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    stock INT,
    supplier_id INT,
    FOREIGN KEY (supplier_id) REFERENCES supplier(supplier_id)
);

desc supplier;
desc inventory;
```

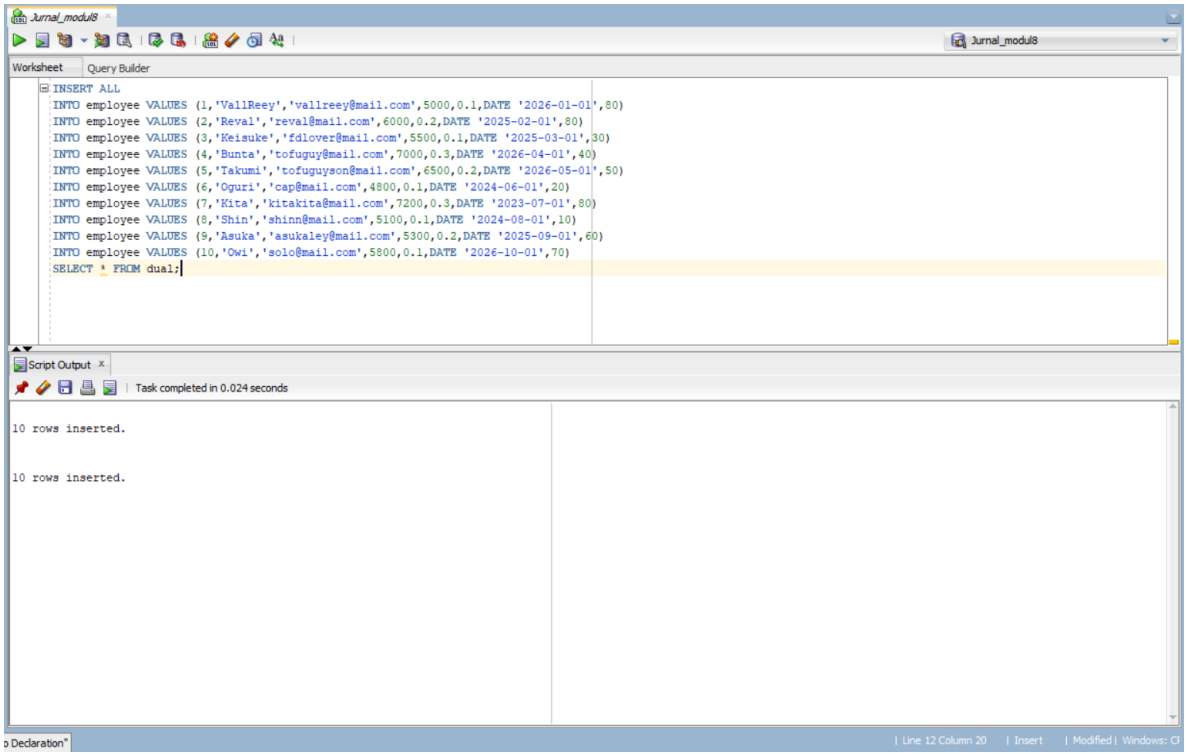
```
Task completed in 0.361 seconds
HIRE_DATE      NOT NULL DATE
DEPARTMENT_ID           NUMBER(38)

Table SUPPLIER created.


Table INVENTORY created.

Name          Null?    Type
------------- -------- -------------
SUPPLIER_ID   NOT NULL NUMBER(38)
SUPPLIER_NAME          VARCHAR2(100)
CONTACT                VARCHAR2(100)
Name          Null?    Type
------------ -------- -------------
PRODUCT_ID   NOT NULL NUMBER(38)
PRODUCT_NAME          VARCHAR2(100)
STOCK                 NUMBER(38)
SUPPLIER_ID           NUMBER(38)
```

2. Analisis table-table dengan compute statistics



3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record)





Jurnal_modul8

Worksheet | Query Builder

```
INSERT ALL
INTO supplier VALUES (1,'Supplier A','0811')
INTO supplier VALUES (2,'Supplier B','0812')
INTO supplier VALUES (3,'Supplier C','0813')
INTO supplier VALUES (4,'Supplier D','0814')
INTO supplier VALUES (5,'Supplier E','0815')
INTO supplier VALUES (6,'Supplier F','0816')
INTO supplier VALUES (7,'Supplier G','0817')
INTO supplier VALUES (8,'Supplier H','0818')
INTO supplier VALUES (9,'Supplier I','0819')
INTO supplier VALUES (10,'Supplier J','0820')
SELECT * FROM dual;
```

Script Output | Task completed in 0.025 seconds

```
10 rows inserted.

10 rows inserted.

10 rows inserted.
```

to Declaration" | Line 12 Column 20 | Insert | Modified | Windows: CI

Jurnal_modul8

Worksheet | Query Builder

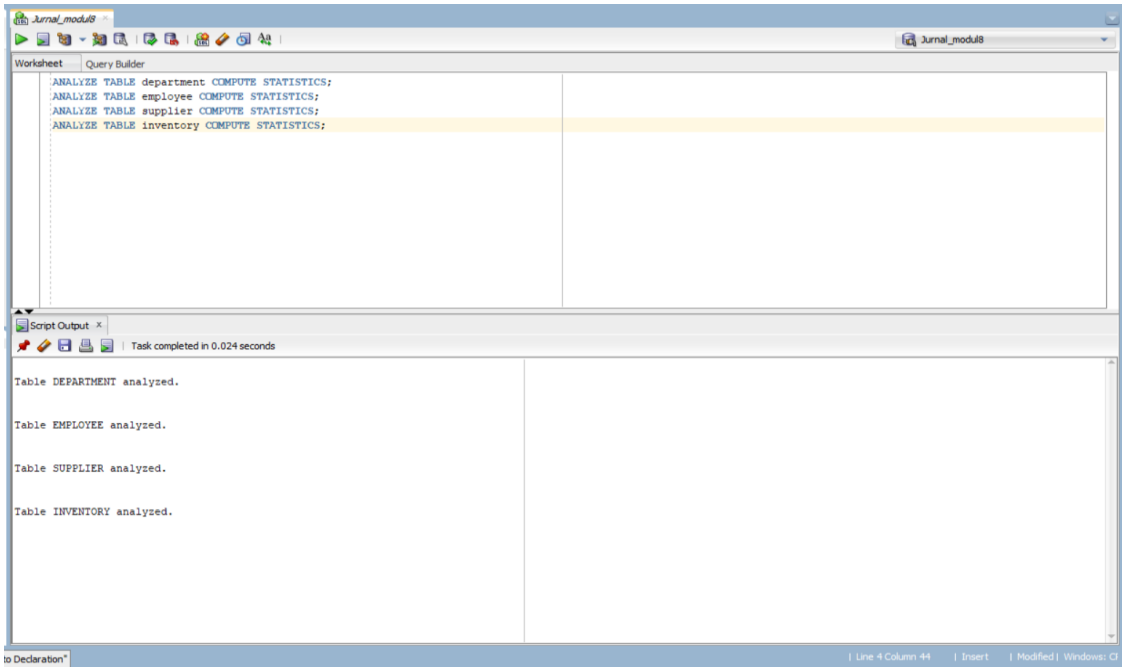
```
INSERT ALL
INTO inventory VALUES (1,'Laptop',10,1)
INTO inventory VALUES (2,'Mouse',50,2)
INTO inventory VALUES (3,'Keyboard',40,3)
INTO inventory VALUES (4,'Monitor',15,4)
INTO inventory VALUES (5,'Printer',8,5)
INTO inventory VALUES (6,'Scanner',6,6)
INTO inventory VALUES (7,'Router',20,7)
INTO inventory VALUES (8,'Switch',25,8)
INTO inventory VALUES (9,'Cable',100,9)
INTO inventory VALUES (10,'UPS',5,10)
SELECT * FROM dual;
```

Script Output | Task completed in 0.033 seconds

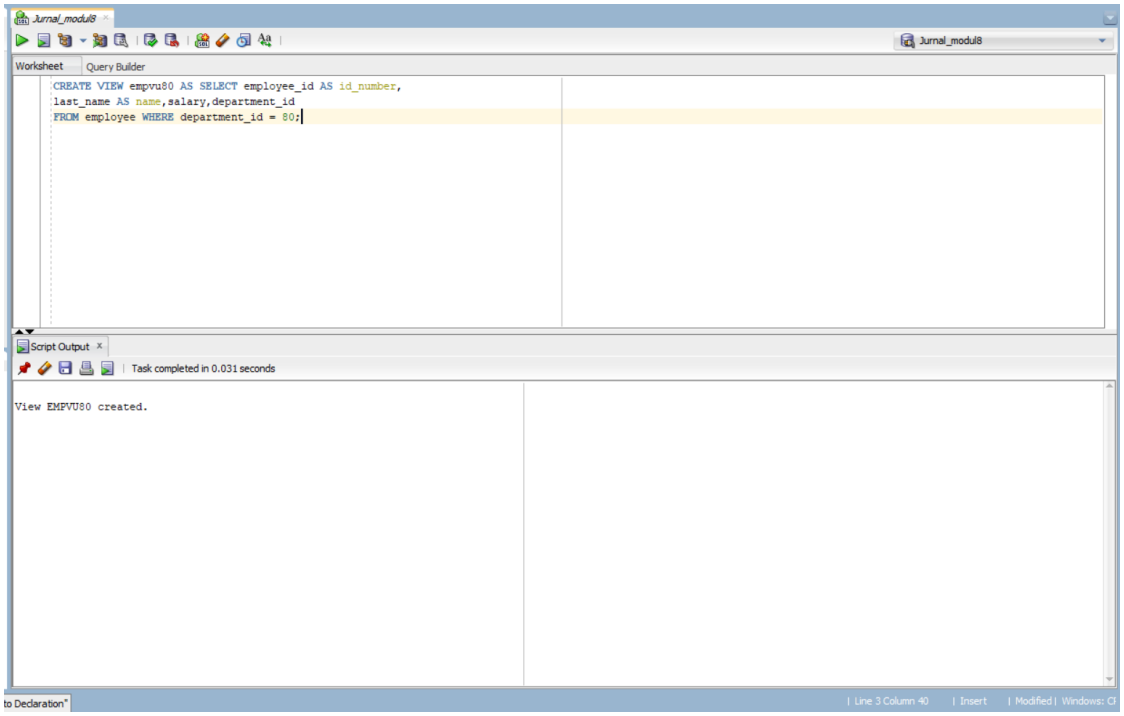
```
10 rows inserted.

10 rows inserted.

10 rows inserted.

10 rows inserted.
```

Declaration" | Line 12 Column 20 | Insert | Modified | Windows: CI

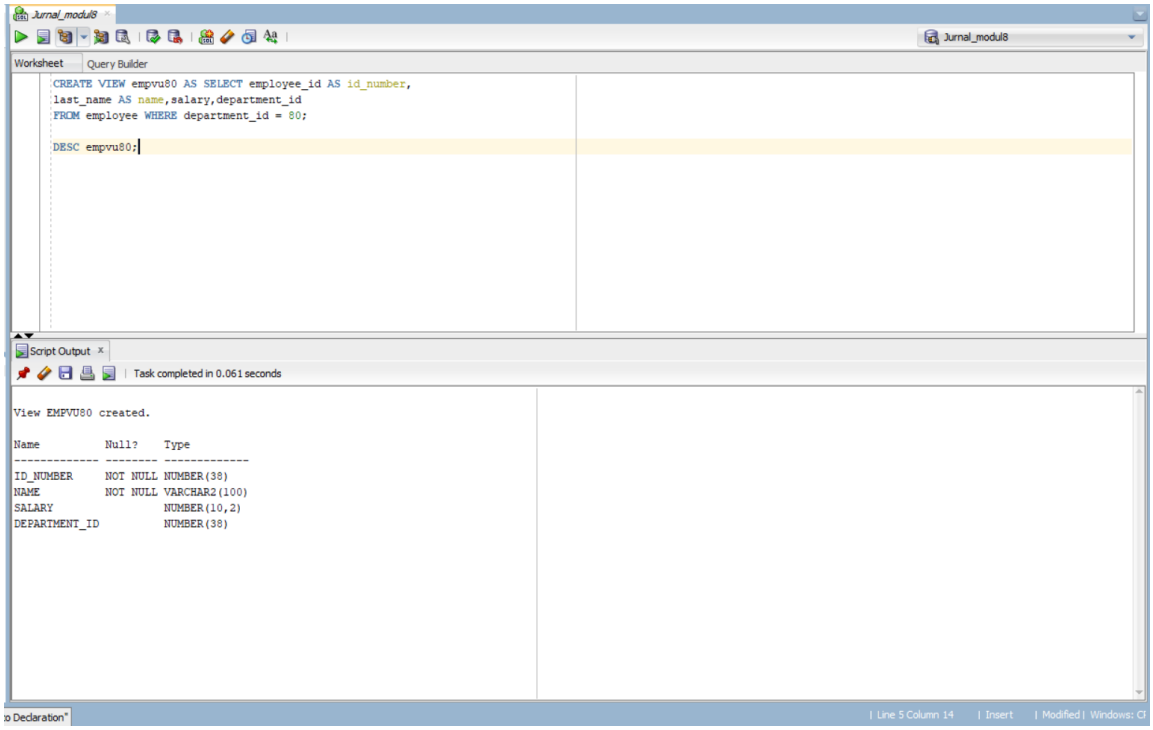
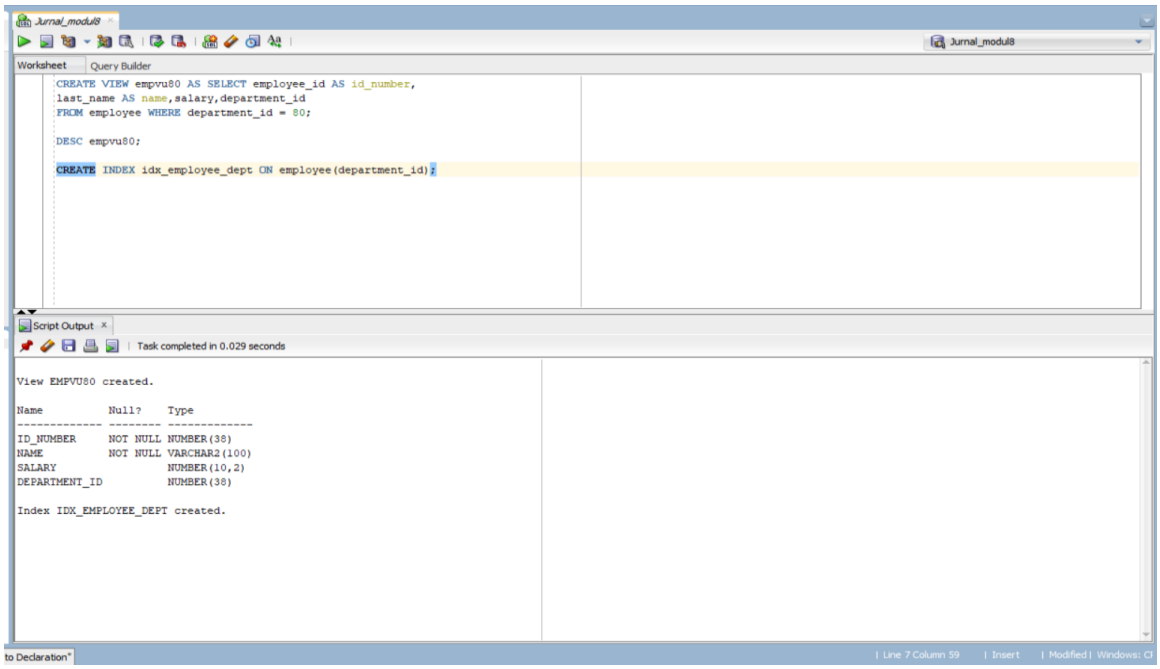
4. Analisis table-table dengan compute statistics



5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80



6. Kemudian tampilkan struktur dari view empvu80.



7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.